

Advanced Topics in Automation and Control Engineering

Aerial Robotics Laboratory

Modeling - Part III

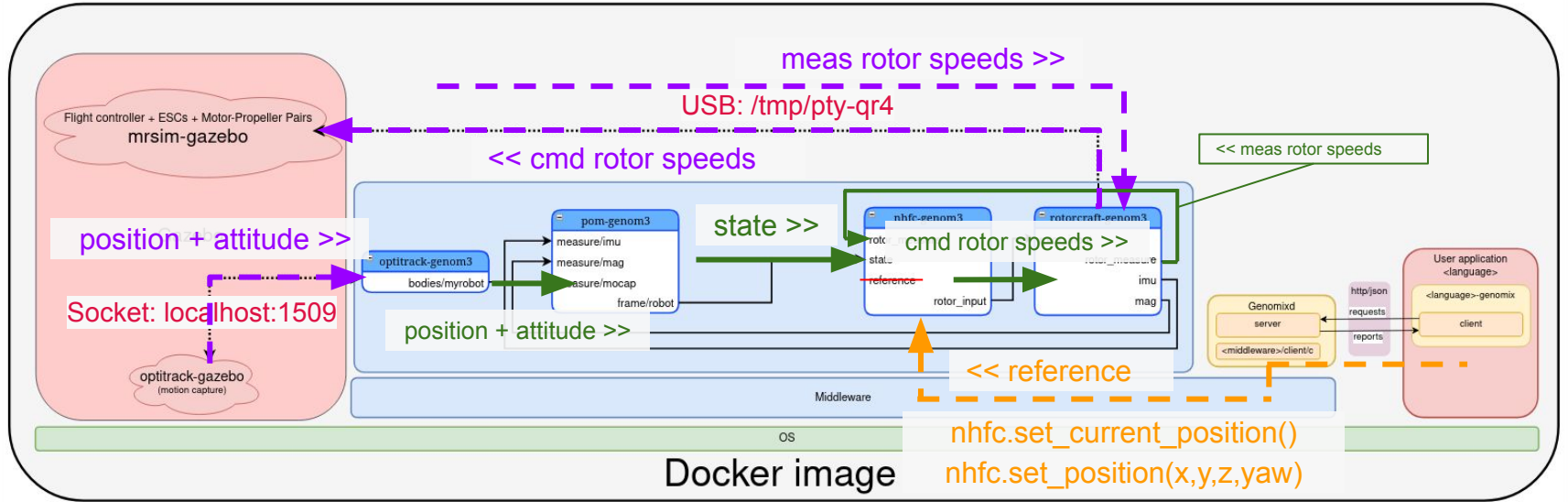
Dr Marco Tognon
Dr Gianluca Corsini



A quick recap first ...

telekyb3 – Quadrotor simulation in Gazebo

Visual representation of the data flow



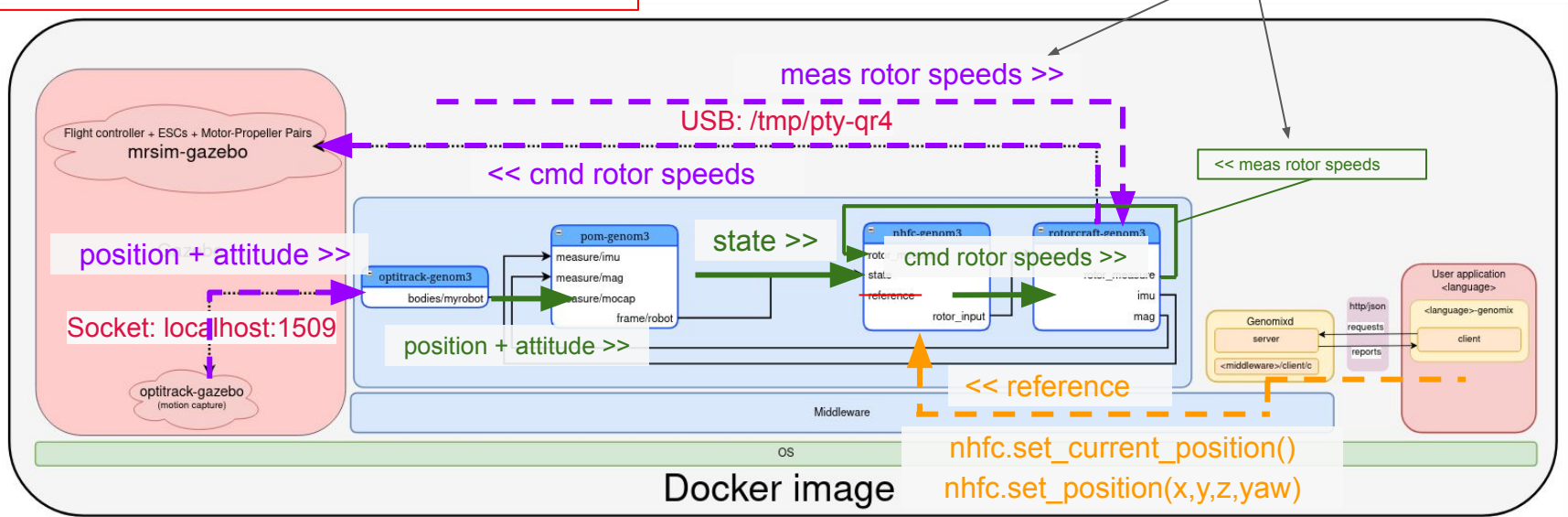
where

- <middleware> = [pocoLibs](#)
- <language> = python ([python-genomix documentation](#))

- *state* := position, attitude, linear and angular velocities, linear and angular accelerations

telekyb3 – Quadrotor simulation in Gazebo

Visual representation of the data flow



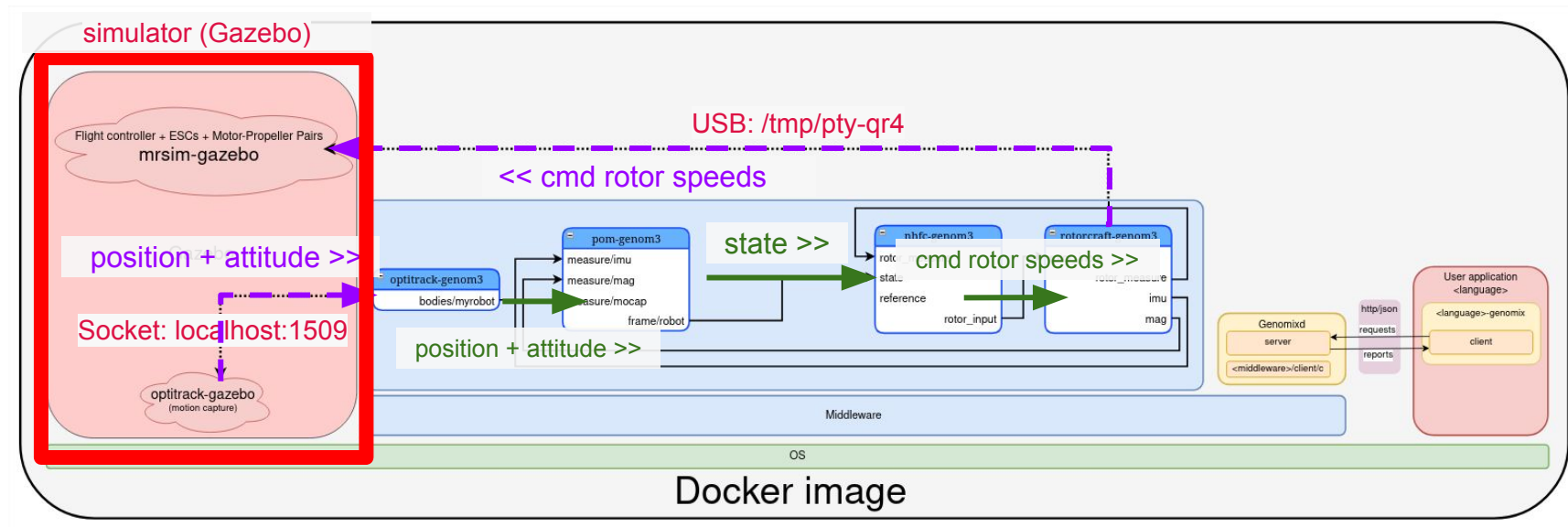
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telekyb3 – Quadrotor simulation in Gazebo

What happens inside Gazebo?



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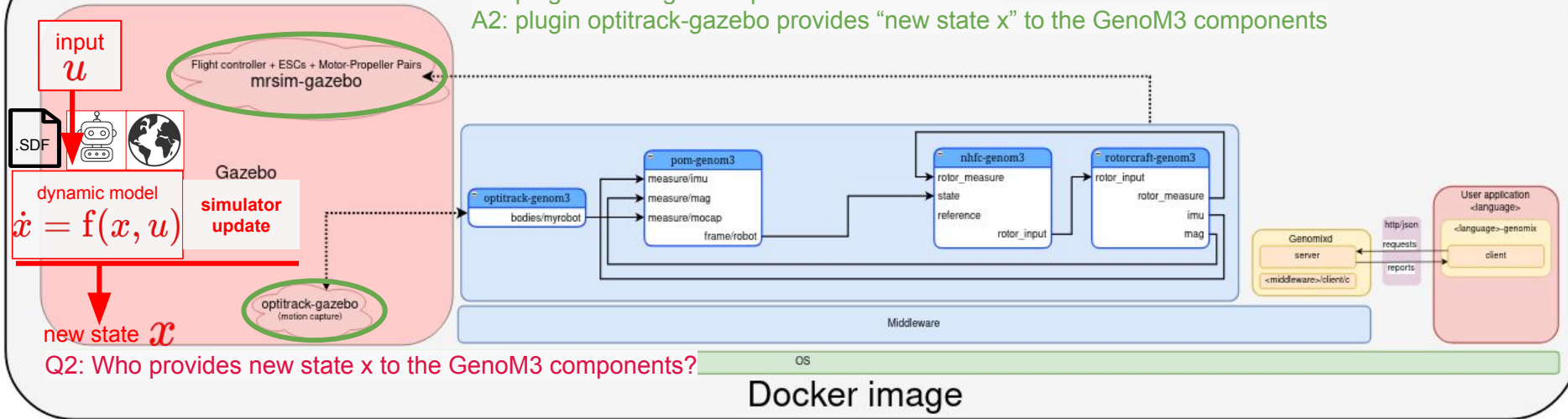
telekyb3 – Quadrotor simulation in Gazebo

What happens inside Gazebo?

Q1: Who provides input u to simulator?

A1: plugin mrsim-gazebo provides u to simulator.

A2: plugin optitrack-gazebo provides “new state x ” to the GenoM3 components



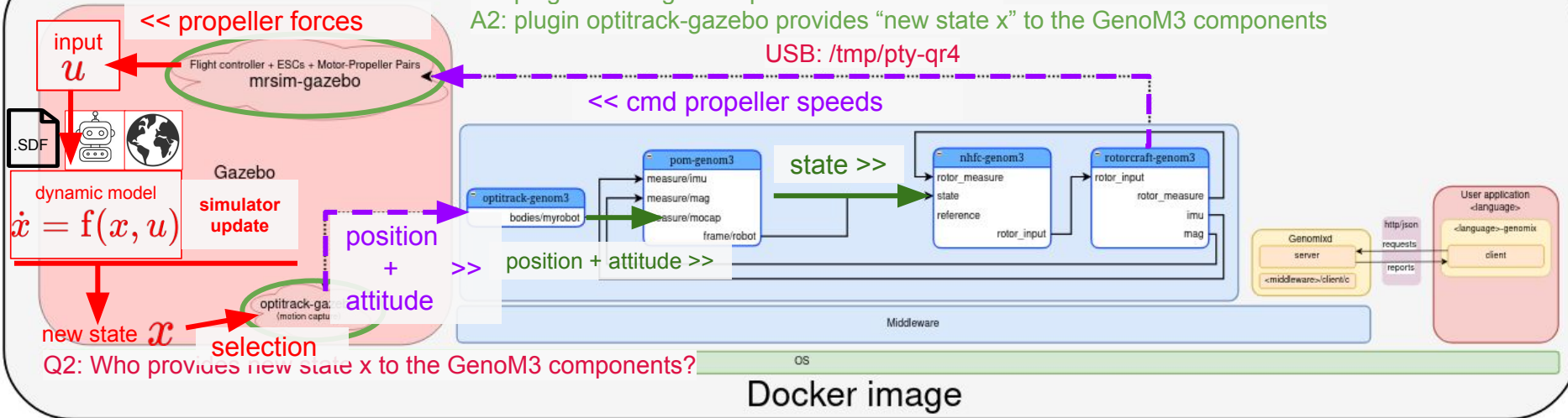
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telekyb3 – Quadrotor simulation in Gazebo

What happens inside Gazebo?

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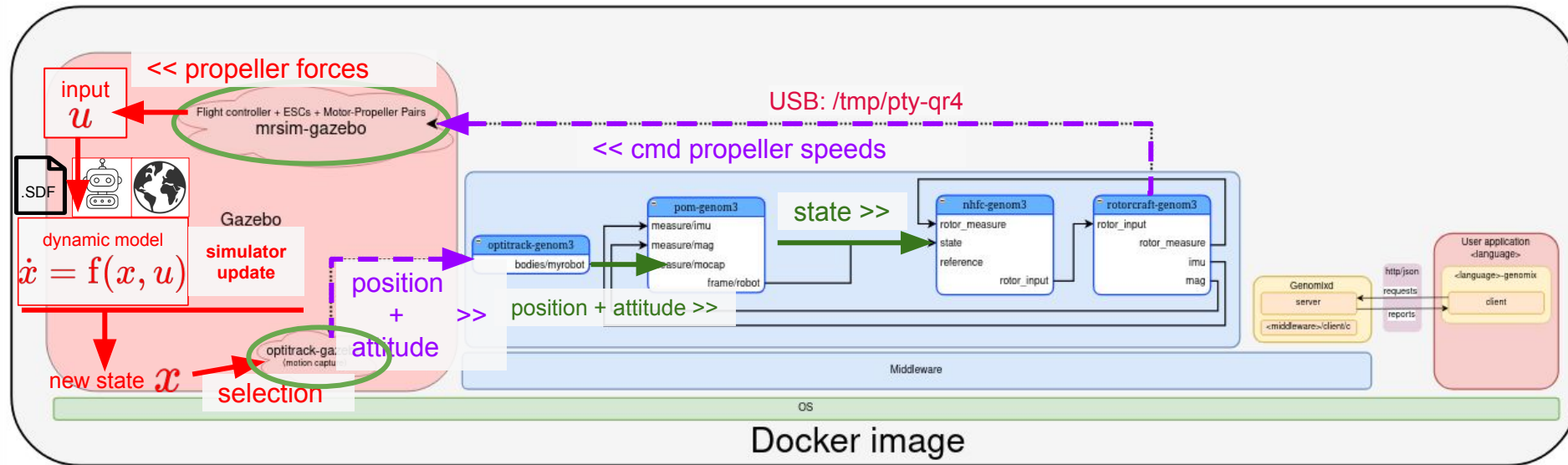


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telekyb3 – Quadrotor simulation in Gazebo

What if we want to **replace Gazebo**?



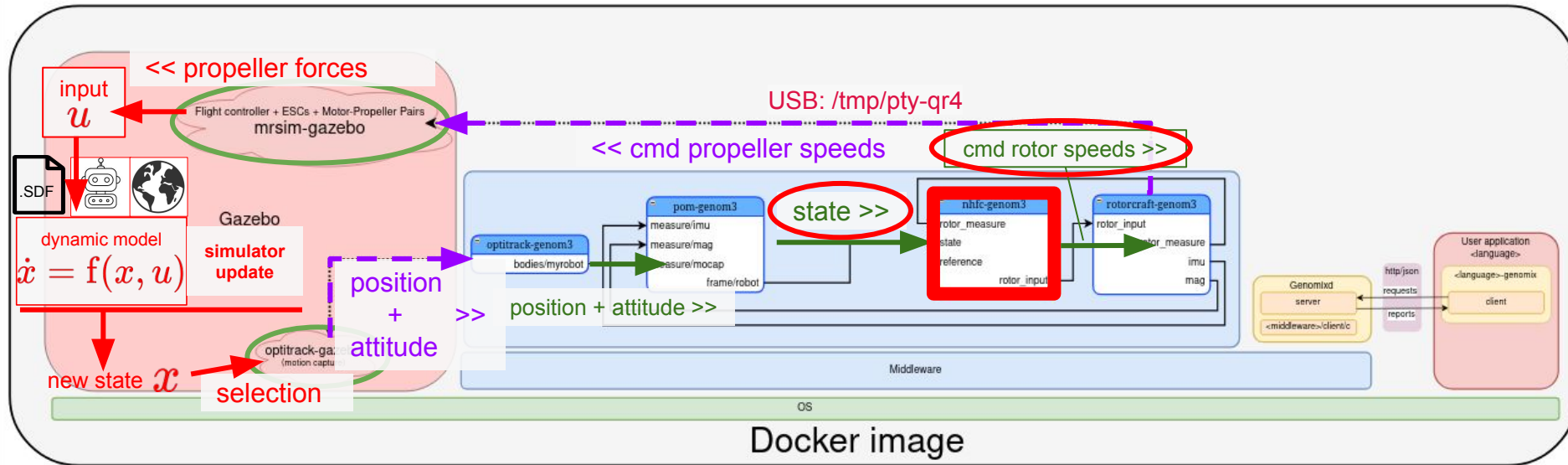
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telekyb3 – Quadrotor simulation in Gazebo

What if we want to replace Gazebo?

Let's consider I/O of the controller (nhfc)



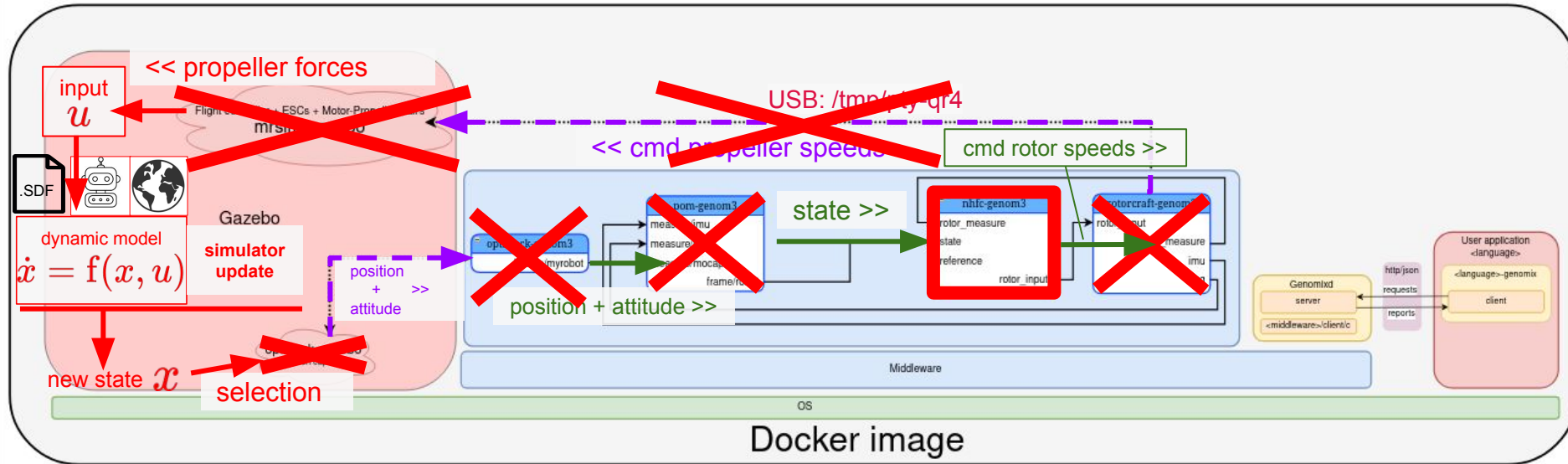
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telekyb3 – Quadrotor simulation in Gazebo

What if we want to replace Gazebo?

Assumption: no hardware-in-the-loop simulation!



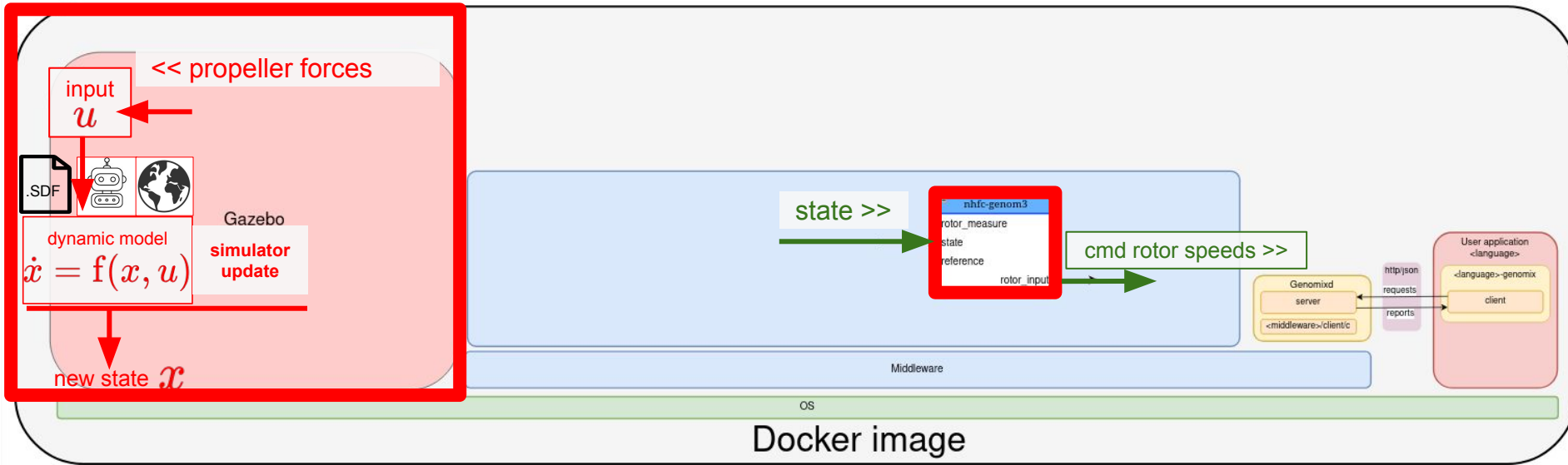
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telekyb3 – Quadrotor simulation in Gazebo

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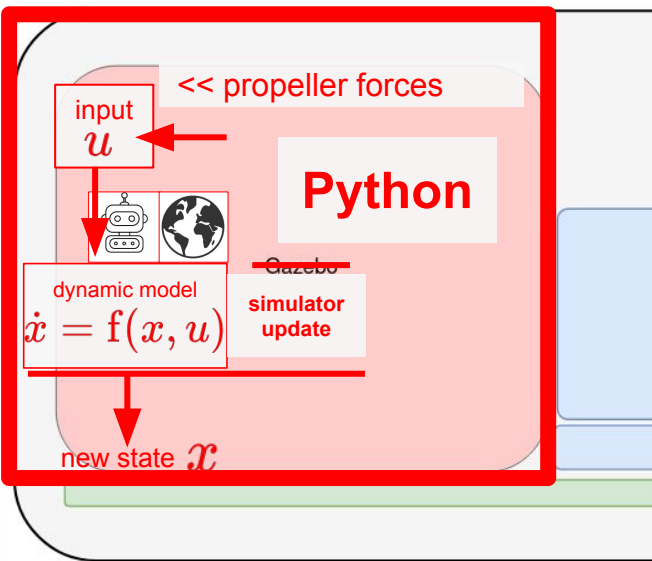
Yesterday assignment

What if we want to replace Gazebo?

Replicate in a Python script the steps performed by the Gazebo simulator engine

Precisely:

1. Initialize simulation with initial state and input conditions
2. Repeat at every time instant $t_k = k\Delta t$:
 - a. Get a new discrete-time input
 - b. Integration of dynamic model with RK4
 - c. Normalize quaternion
 - d. Compute state at new time instant
 - e. Check robot height:
if necessary, then apply ground reaction



where

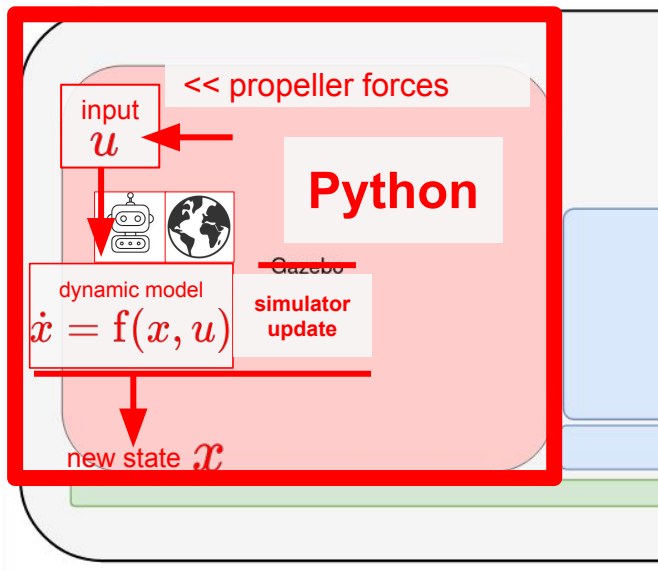
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Yesterday assignment

What if we want to replace Gazebo?

Considerations:

- Create an independent Python script!
- Define $\Delta t = 1\text{ms}$
- Test simulator with
 - lifting force equal to weight force
 - lifting force 1N above weight force
- No python-genomix
- No GenoM3 components
- Generate plots as yesterday assignment
 - No controller setpoints here!
 - Only measured (current) robot state!



Yesterday assignment – Challenges

1. How to derive $f(x,u)$?
 - a. Aerial robot modeling + Newton-Euler formalism
 - b. State-space modelisation
2. How to discretize $f(x,u)$ (time-continuous function) and numerically integrate it?
 - a. Compute x_{k+1} given x_k and $u_k \rightarrow$ computer-friendly

Yesterday assignment – Challenge #1

$$\mathbf{x} = \begin{bmatrix} \overset{1\ 2\ 3}{W \mathbf{p}_B^T} & \overset{4\ 5\ 6\ 7}{W \mathbf{q}_B^T} & \overset{8\ 9\ 10}{B \mathbf{v}_B^T} & \overset{11\ 12\ 13}{B \boldsymbol{\omega}_B^T} \end{bmatrix}^T$$

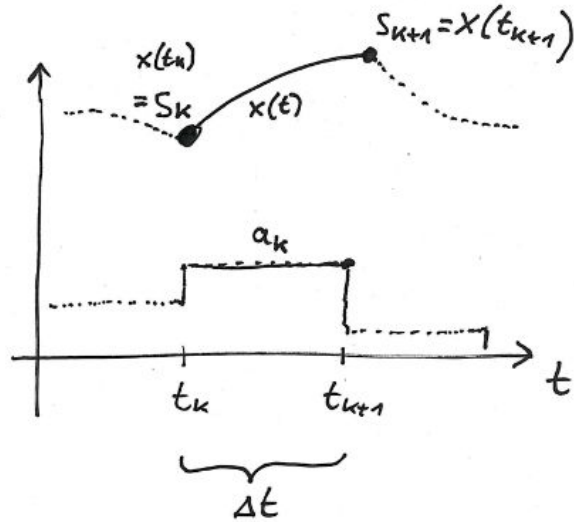
$$\mathbf{u} = \mathbf{u}_\lambda$$

scalar first!

$$\dot{\mathbf{x}} = \begin{bmatrix} W \dot{\mathbf{p}}_B \\ W \dot{\mathbf{q}}_B \\ W \dot{\mathbf{v}}_B \\ B \dot{\boldsymbol{\omega}}_B \end{bmatrix} = \begin{bmatrix} W \dot{\mathbf{p}}_B \\ W \dot{\mathbf{q}}_B \\ W \ddot{\mathbf{p}}_B \\ B \dot{\boldsymbol{\omega}}_B \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{8:10} \\ ? \\ \tilde{\mathbf{f}}_{1:3} \\ \tilde{\mathbf{f}}_{4:6} \end{bmatrix} = \begin{bmatrix} \overset{\mathbf{x}_{8:10}}{\frac{1}{2} \begin{bmatrix} 0 \\ W \boldsymbol{\omega}_B \end{bmatrix} \otimes \begin{bmatrix} \tilde{\mathbf{f}}_{1:3} \\ \tilde{\mathbf{f}}_{4:6} \end{bmatrix}} \\ W \mathbf{q}_B \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{8:10} \\ (\mathbf{x}_{11:13}, \mathbf{x}_{4:7}) \\ \tilde{\mathbf{f}}_{1:3} \\ \tilde{\mathbf{f}}_{4:6} \end{bmatrix} = \boxed{\mathbf{f}(\mathbf{x}, \mathbf{u})} \quad \#1$$

Yesterday assignment – Challenge #2

Runge Kutta (RK) methods



Intermediate updates:

$$k_1 = f(x_k, u_k)$$

$$k_2 = f\left(x_k + \frac{\Delta t}{2} k_1, u_k\right)$$

$$k_3 = f\left(x_k + \frac{\Delta t}{2} k_2, u_k\right)$$

$$k_4 = f(x_k + \Delta t k_3, u_k)$$

Then:

$$\boxed{x_{k+1}} = x_k + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad \text{RK 4-th order}$$

#2

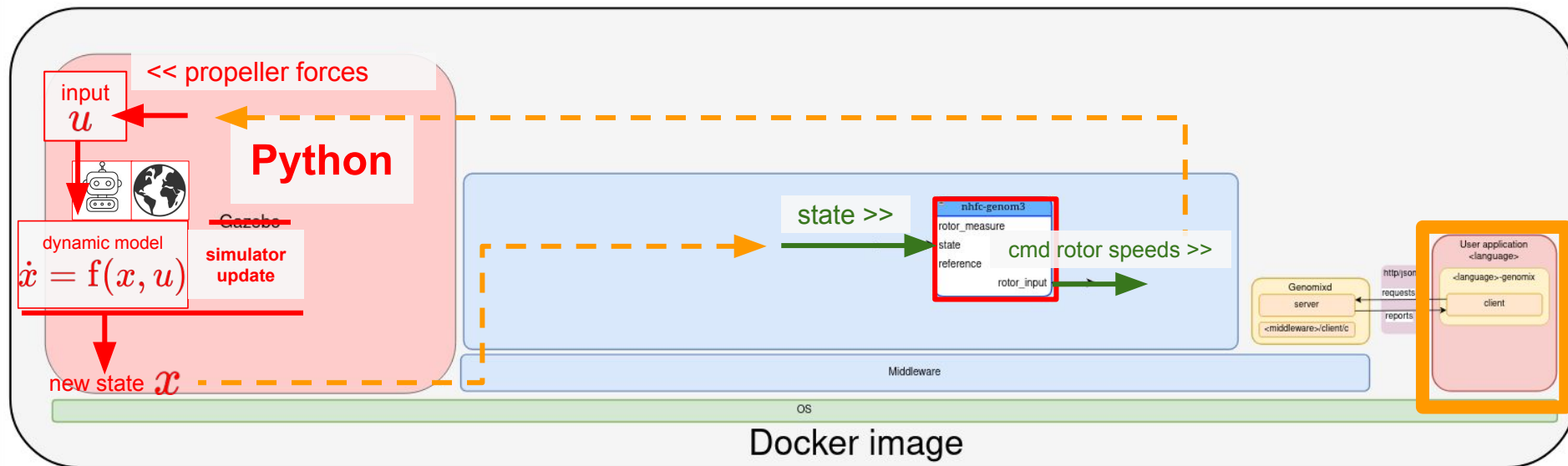
Reference: J. Boedecker and M. Diehl, University Freiburg, [Model Predictive Control and Reinforcement Learning – Lecture 2: Dynamic Systems and Simulation](#).



Today...

Today assignment

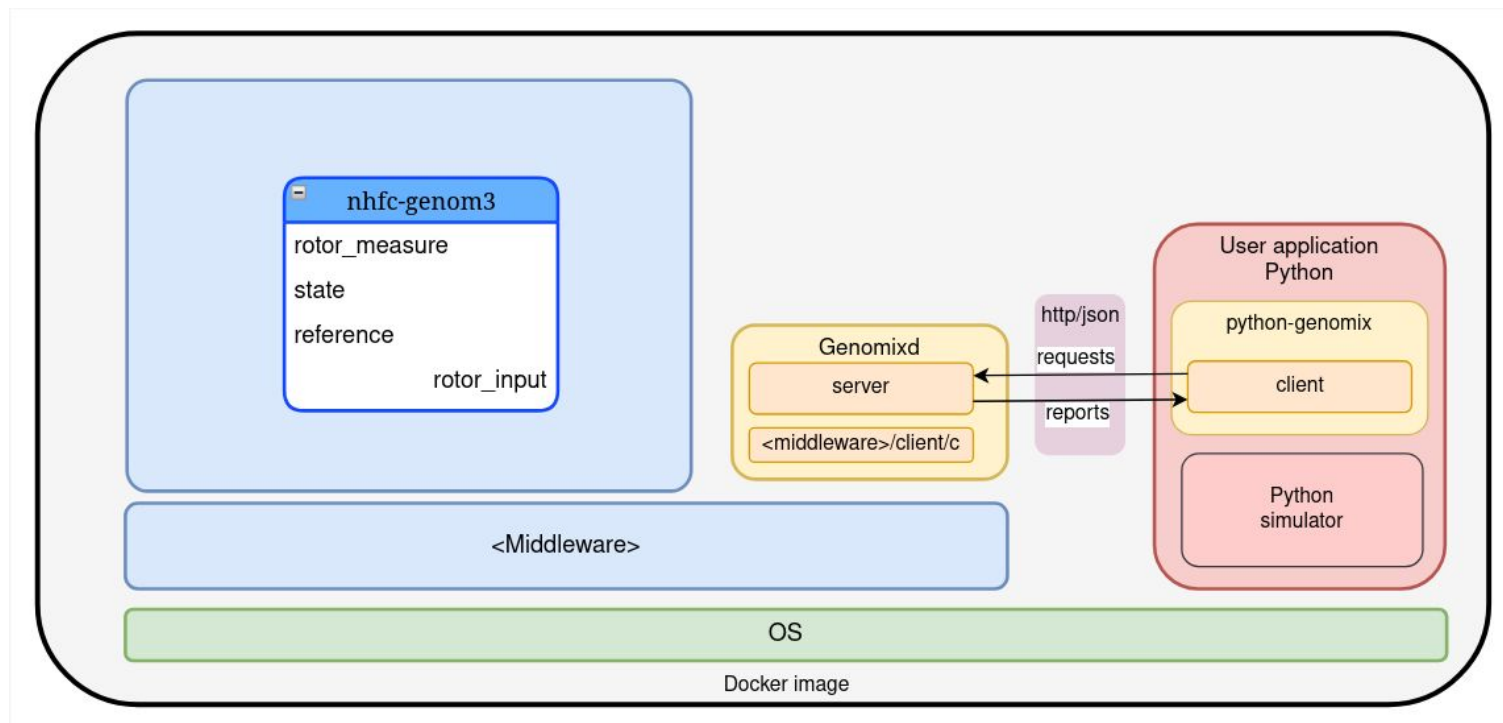
How? Embed our Python simulator in an user application and use python-genomix for the communication!



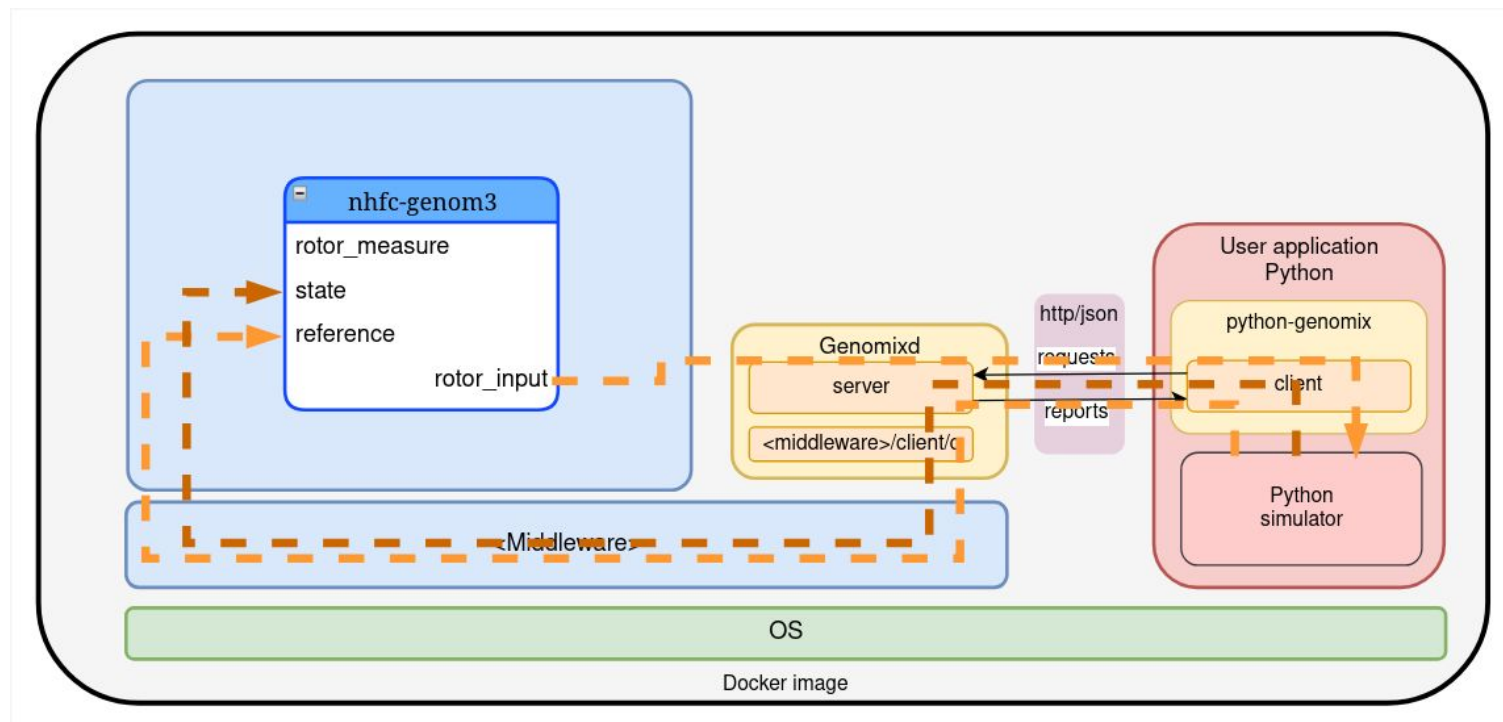
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Today assignment



Today assignment



How to connect our simulator to telekyb3?

Our simulator is just a Python script and it is not connected to telekyb3 (GenoM3 components, ...)

How to perform this connection?

Use python-genomix:

- Call services
- Read data from ports
- Write data to ports

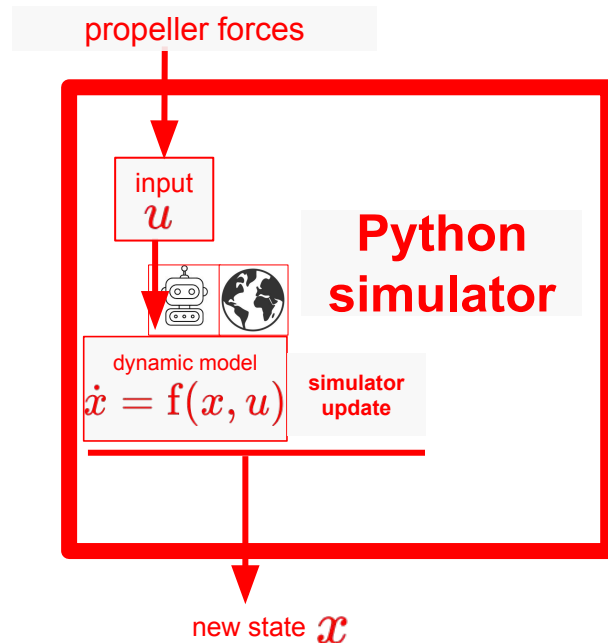
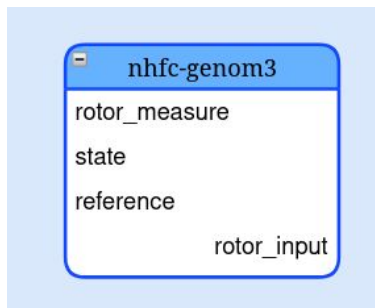
How to connect our simulator to telekyb3?

Missing steps:

1. Provide reference to controller
2. Obtain propeller speeds (forces) from the controller
3. Provide state to controller

Remember:

- No hardware-in-the-loop
 - Perfect state estimation
- Neglect rotor_measure
 - Not needed here

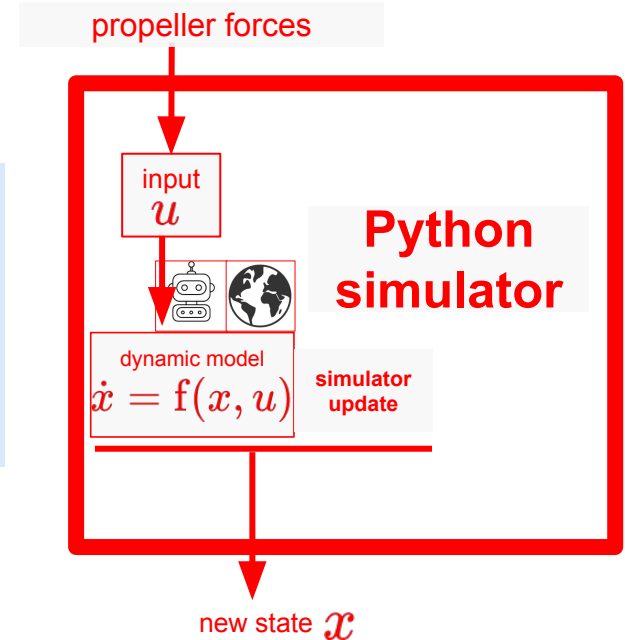
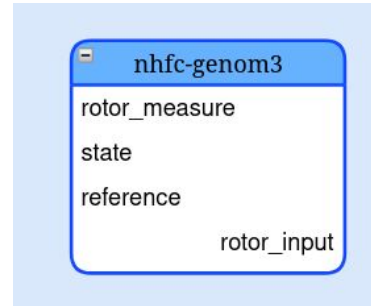


1. Provide reference to controller

How?

Use services 'set_current_position()' and 'set_position(x,y,z,yaw)'!

Nothing new here :)



2. Obtain propeller speeds (forces)

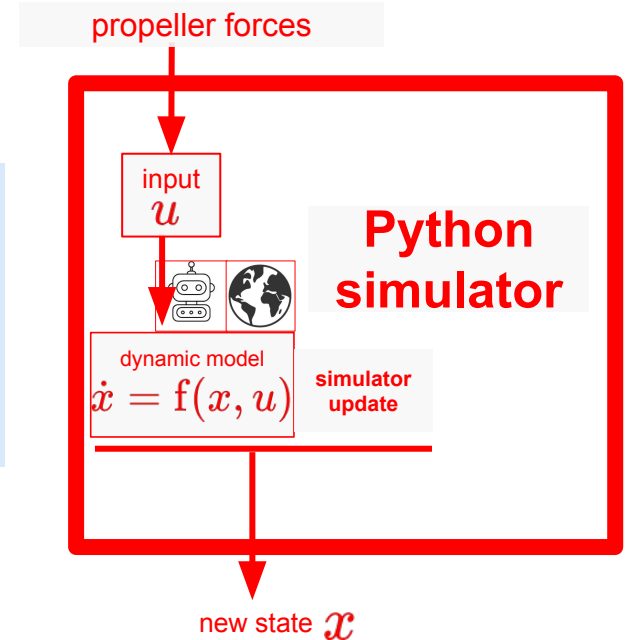
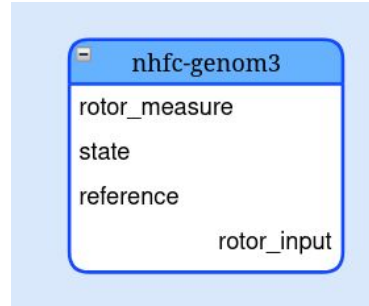
How?

Read data from output port 'rotor_input' of GenoM3 component 'nhfc-genom3'.

No service available to get that data!

New problem:

#1 How to read data from output ports in python-genomix?



3. Provide state to controller

How?

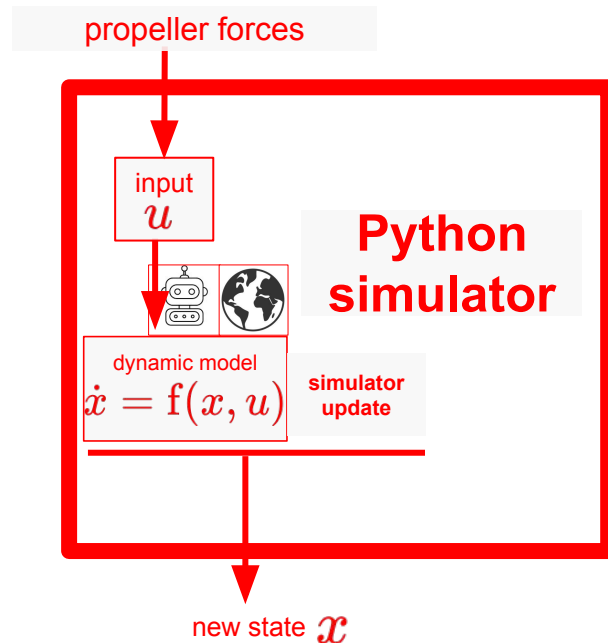
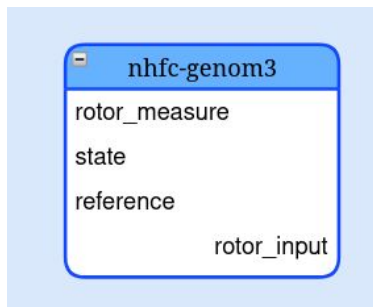
Write data to input port 'state' of
GenoM3 component 'nhfc-genom3'.

No service available to do that!

NB: the service 'set_state' is for *desired*
state (and not for state feedback)!

New problem:

#2 How to write data to input ports in
python-genomix?



python-genomix – Read data from output port

```
>>> import genomix

>>> client = genomix.connect()

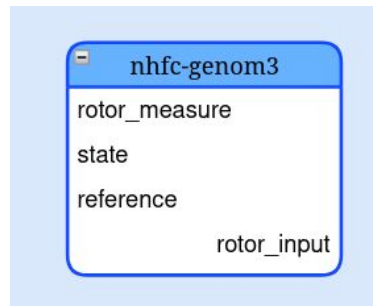
>>> nhfc = client.load("nhfc")

>>> nhfc.rotor_input()

{'rotor_input': {'ts': {'sec': 1776335485, 'nsec':
732712000}, 'control': '::or_rotorcraft::velocity',
'desired': [72.01208962143923, 67.35095894473037,
72.7424517916212, 65.3161164045903]}}
```

returned value is a dictionary!

Example: nhfc-genom3



<https://git.openrobots.org/projects/python-genomix/pages/README#reading-output-ports>

python-genomix – Write data to input port

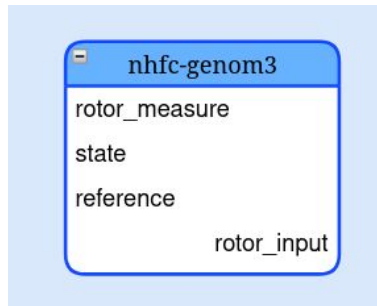
```
>>> import genomix  
  
>>> client = genomix.connect()  
  
>>> nhfc = client.load("nhfc")  
  
>>> state_port = nhfc.state('my_state')  
  
>>> nhfc.connect_port({ 'local': 'state',  
                        'remote': 'my_state' })
```

write data to the input port:

```
>>> state_port data definition??
```

Determine message type of a port!

Example: nhfc-genom3



<https://git.openrobots.org/projects/python-genomix/pages/README#publishing-input-ports>

Determine message type of a port

From the documentation of [nhfc-genom3](#):

rotor_input (out)

Data structure

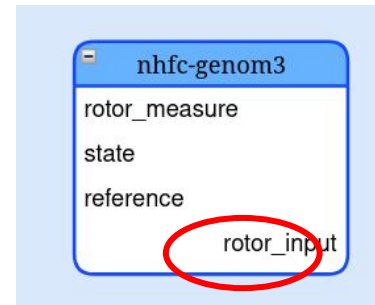
- struct ::or_rotorcraft::input rotor_input
 - struct ::or::time::ts ts
 - long sec
 - long nsec
 - enum ::or_rotorcraft::control_type control ∈ { velocity, throttle }
 - sequence< double, 8 > desired

Q1: Components use data structures (usually) in C/C++ (e.g. 'struct'), how to convert them to **Python data types**?

Refer to python-genomix documentation

<https://git.openrobots.org/projects/python-genomix/pages/README#mapping-between-idl-types-and-python-arguments>

Example: nhfc-genom3



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```

returned value is a dictionary!

Determine message type of a port

From the documentation of [nhfc-genom3](#):

rotor_input (out)

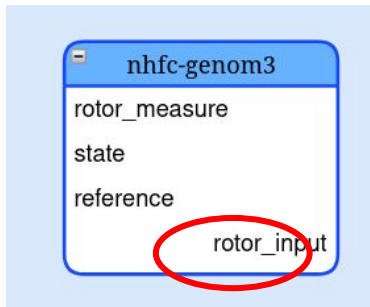
Data structure

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 - struct ::or::time::ts ts
 - long sec
 - long nsec
 - enum ::or_rotorcraft::control_type control ∈ { velocity, throttle }
 - sequence< double, 8 > desired

**Q2: Where are defined the types as
`::or\_rotorcraft\_input`, `::or::time::ts`, etc?**

Refer to openrobots-idl <https://git.openrobots.org/projects/openrobots-idl/repository>

Example: nhfc-genom3



```
>>> import genomix

>>> client = genomix.connect()

>>> nhfc = client.load("nhfc")

>>> nhfc.rotor_input()

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```

returned value is a dictionary!

openrobots-idl

IDL stands for *Interface Definition Language*, i.e., a descriptive, language-neutral syntax used to define data types.

Indeed, in reality, GenoM3 components use data types in a language-neutral syntax, and usually they are converted into C/C++!

For reference, refer to

<https://git.openrobots.org/projects/genom3/pages/mappings/index>.

For example, for the conversion of basic data types in C, refer to the subpage

https://git.openrobots.org/projects/genom3/pages/mappings/c#basic_type.

openrobots-idl – What to really retain

1. Look for the data type definition in openrobots-idl

<https://git.openrobots.org/projects/openrobots-idl/repository>

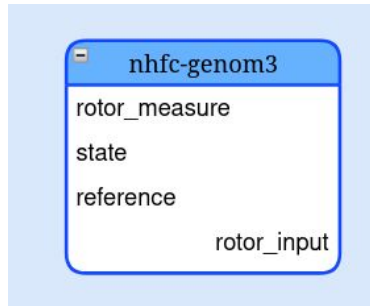
2. Apply the mapping in

<https://git.openrobots.org/projects/python-genomix/pages/README#mappings>

Live example for nhfc's output port 'rotor_input'

- nhfc → output port [rotor_input](#) → port message type
'struct ::or_rotorcraft::input'
- [openrobots-idl](#) → [rotorcraft-genom3](#)
- [python-genomix mappings](#)

rotor_input (out)



Data structure

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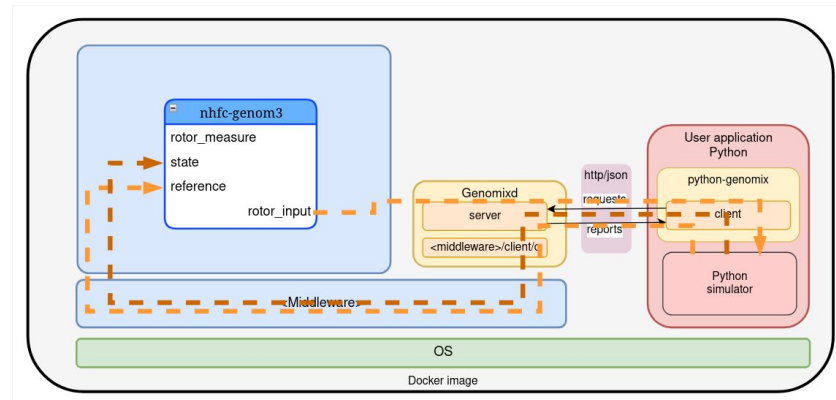
What about input port 'state'?

Today assignment

Replicate in a Python script the steps performed by the Gazebo simulator engine and connect it to 'nhfc-genom3'

Precisely:

1. Initialize simulation with initial state and input conditions
2. Repeat at every time instant $t_k = k\Delta t$:
 - a. Get a new discrete-time input from 'nhfc-genom3' reading output port 'rotor_input'
 - b. Integration of dynamic model with RK4
 - c. Normalize quaternion
 - d. Compute state at new time instant
 - e. Check robot height → if necessary, then apply ground reaction
 - f. Provide state to 'nhfc-genom3' at input port 'state'



Use the provided Python script template!